

Customer Retention & Churn Analysis Report

Future Interns – Data Science & Analytics Task 2

1. Project Overview

Objective

The objective of this project is to analyze customer churn behavior and identify the factors influencing customer retention. The analysis helps businesses understand why customers leave and what strategies can improve retention and reduce churn.

2. Dataset Information

Dataset Used

Telco Customer Churn Dataset (Kaggle)

Dataset Description

The dataset contains customer demographic information, subscription details, internet services, payment methods, tenure, monthly charges, and churn status. It is widely used for customer retention and churn analysis.

3. Tools Used

- Microsoft Power BI
 - Microsoft Excel
-

4. Data Preparation

The following data preparation steps were performed:

- Imported the dataset into Power BI.
- Verified and corrected data types.
- Converted TotalCharges to numeric format.
- Handled missing values.
- Created DAX measures for Total Customers, Churned Customers, Active Customers, and Churn Rate.
- Created Tenure Groups for customer lifetime analysis.

5. Dashboard Analysis

Key Metrics

- Total Customers: 3000
- Churned Customers: 939
- Active Customers: 3000
- Churn Rate: 26.92%

6. Key Insights

Insight 1: Overall Churn Rate

The overall customer churn rate is 26.92%, indicating that approximately one in four customers leaves the service.

Insight 2: Contract Type Impact

Customers with Month-to-Month contracts show significantly higher churn than customers with One-Year or Two-Year contracts.

Insight 3: Customer Lifetime Trends

Most customer churn occurs within the first 12 months of service, highlighting the importance of early customer engagement and support.

Insight 4: Payment Method Analysis

Customers using Electronic Check as their payment method show higher churn compared to customers using other payment methods.

Insight 5: Internet Service Analysis

Fiber Optic customers represent a large proportion of churned customers compared to other internet service categories.

7. Retention Drivers

The analysis identified the following key factors affecting customer retention:

- Contract Type
- Customer Tenure
- Payment Method
- Internet Service Type
- Monthly Charges

8. Business Recommendations

Recommendation 1

Encourage customers to switch from Month-to-Month contracts to longer-term plans through discounts and incentives.

Recommendation 2

Improve onboarding and customer support during the first year of service.

Recommendation 3

Implement targeted retention campaigns for customers identified as high-risk segments.

Recommendation 4

Provide personalized offers and loyalty benefits for customers using Electronic Check payment methods.

Recommendation 5

Develop long-term loyalty programs to improve customer retention and customer lifetime value.

9. Limitations

The Telco Customer Churn dataset does not contain customer signup dates or regional cohort information. Therefore, traditional cohort analysis by signup month or region could not be performed. Instead, tenure-based segmentation was used to analyze customer lifetime behavior and retention trends.

10. Conclusion

This analysis demonstrates that customer churn is strongly influenced by contract type, tenure, payment method, and internet service selection. By implementing targeted retention strategies and improving customer engagement, organizations can reduce churn, improve customer satisfaction, and increase long-term revenue growth.

11. Dashboard Screenshot

